



**Six diagnostics → outcomes in this study (n = 5, tie-aware exact permutation)**

|                                         |                                                                            |
|-----------------------------------------|----------------------------------------------------------------------------|
| <b>D1 Windowed estimator variance</b>   | noise $\geq$ signal (0.3–1.2 $\times$ ): ordering unresolvable at 26 ns    |
| <b>D2 Estimator–consistency matrix</b>  | no cell significant (all $p \geq 0.27$ ); sign unstable across estimators  |
| <b>D3 Convergence–timescale scaling</b> | no decay over 2–24 ns windows; no extrapolated sampling length             |
| <b>D4 Ordinal sensitivity model</b>     | no decision-level cell ( $P \leq 0.78$ across assumed bands)               |
| <b>D5 Counterfactual sensitivity</b>    | leave-one-out $-0.63 \dots +0.63$ ; single-point exclusion flips sign      |
| <b>D6 Chemical-state audit</b>          | protonation corrections move $p -0.67 \rightarrow -0.05$ : state dominates |

**Verdict:** no robust association between the computed spans and trapping potency — not established under any chemically defensible ligand-state panel. The transferable output is the audit workflow itself.